

AAAA

Your Name

October 6, 2026

Abstract

1 Introduction

I had browsed through existing works in recent 3 years in the field of Intelligent Vehicle and summarized the topic and the trend of the research. [1–12]

There are several main problems in the field of Intelligent Vehicle, including bandwidth & communication bottlenecks, latency & temporal asynchrony, heterogeneity & cross-model compatibility, infrastructure-assisted efficiency (V2I), robustness, security & reliability, and sim-to-real gap & empirical deployment constraints.

Then I focus on the problem of latency & temporal asynchrony in cooperative perception. These two challenges are critical to spatial-temporal data consistency. Latency refers to the delay from sensation to fusion and decision-making, which can be caused by network transmission, data processing, and algorithmic computation. This latency comes from bottlenecks in communication, computation, and synchronization, which can lead to outdated or incomplete information being used for decision-making. Temporal asynchrony refers to the misalignment of data timestamps from different sources, including RSUs and CAVs, which can lead to inconsistencies in the perception results.

Usually, data are fused trivially by addition or concatenation, which can lead to spatiotemporal misalignment and inaccurate perception results. Moving objects are predicted to be in wrong historical positions instead of their current positions, leading to phantom detections, tracking fragmentation, and dangerous miscalculations in downstream motion planning.

And nowadays, there are several methods to address the latency and temporal asynchrony problem in cooperative perception, each compensating from different aspects. [13–18]

These include predicting current features using historical information, performing spatial warping of features via motion information or BEV flow, and leveraging trajectory information to establish correspondences across different timestamps. However, these methods align features in a single frame, which may not fully capture the temporal dynamics of moving objects and may not be robust to occlusions or missing data. Therefore, recently research moves toward object level.

2 Object level Alignment

References

- [1] Z. Lei, S. Ren, Y. Hu, W. Zhang, and S. Chen, “Latency-aware collaborative perception,” in *European Conference on Computer Vision (ECCV)*, 2022. [Online]. Available: <https://arxiv.org/abs/2207.08560>
- [2] Y. Hu, S. Fang, Z. Lei, Y. Zhong, and S. Chen, “Where2comm: Communication-efficient collaborative perception via spatial confidence maps,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. [Online]. Available: <https://arxiv.org/abs/2209.12836>
- [3] Y. Xia, Q. Yuan, G. Luo, X. Fu, Y. Li, X. Zhu, T. Luo, S. Chen, and J. Li, “One is plenty: A polymorphic feature interpreter for immutable heterogeneous collaborative perception,” *arXiv preprint arXiv:2411.16799*, 2024. [Online]. Available: <https://arxiv.org/abs/2411.16799>
- [4] H. Si, E. Javanmardi, and M. Tsukada, “I don’t understand, but if we train together, i will: Progressive heterogeneous collaborative perception,” *arXiv preprint arXiv:2509.09310*, 2025. [Online]. Available: <https://arxiv.org/abs/2509.09310>
- [5] J. Wang, Z. Gao, and T. Nordström, “Communication efficient cooperative perception via codebook-free vector quantization,” *IEEE Access*, 2026. [Online]. Available: <https://doi.org/10.1109/ACCESS.2026.3674083>
- [6] C. Wei, G. Wu, and M. J. Barth, “Cooperative perception for automated driving: A survey of algorithms, applications, and future directions,” *Proceedings of the IEEE*, 2025. [Online]. Available: <https://doi.org/10.1109/JPROC.2025.3608874>
- [7] J. Wang and T. Nordström, “Latency robust cooperative perception using asynchronous feature fusion,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2025. [Online]. Available: <https://doi.org/10.1109/WACV61041.2025.00476>
- [8] A. Poibrenski, F. Nozarian, F. Rezaeianaran, and C. Müller, “Robust collaborative perception: Combining adversarial training with consensus mechanism for enhanced v2x security,” in *IEEE Intelligent Vehicles Symposium (IV)*, 2025. [Online]. Available: <https://doi.org/10.1109/IV64158.2025.11097632>
- [9] Y. Gao, Y. Yang, J. Lv, X. He, and S. Chen, “Spatiotemporal differential scheduling for vehicle-to-infrastructure cooperative perception,” *IEEE Transactions on Vehicular Technology*, 2025. [Online]. Available: <https://doi.org/10.1109/WCNCW67598.2026.11555357>
- [10] J. Wang and T. Nordström, “Towards model-agnostic cooperative perception,” in *International Joint Conference on Neural Networks (IJCNN)*, 2025. [Online]. Available: <https://doi.org/10.1109/IJCNN64981.2025.11229412>
- [11] T. Huang, J. Liu, X. Zhou, D. C. Nguyen, M. R. Azghadi, Y. Xia, Q.-L. Han, and S. Sun, “Vehicle-to-everything cooperative perception for

- autonomous driving,” *Proceedings of the IEEE*, 2025. [Online]. Available: <https://doi.org/10.1109/JPROC.2025.3600903>
- [12] A. Guo, S. Zhang, E. Tang, X. Gao, H. Pang, H. Tian, Y. Mu, W. Wen, C. Fang, and Z. Chen, “When autonomous vehicle meets v2x cooperative perception: How far are we?” *arXiv preprint arXiv:2501.00000*, 2025. [Online]. Available: <https://arxiv.org/abs/2509.24927>
 - [13] S. Wei, Y. Wei, Y. Hu, Y. Lu, Y. Zhong, S. Chen, and Y. Zhang, “Asynchrony-robust collaborative perception via bird’s eye view flow,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
 - [14] Z. Song, L. Yang, F. Wen, and J. Li, “Traf-align: Trajectory-aware feature alignment for asynchronous multi-agent perception,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
 - [15] S. Du, S. Qu, T. Wang, X. Zhang, Y. Zhu, J. Mao, F. Lu, Q. Lin, and G. Chen, “Rcp-bench: Benchmarking robustness for collaborative perception under diverse corruptions,” in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
 - [16] M. Yazgan, A. X. Arasan, and J. M. Zöllner, “Efficomm: Bandwidth efficient multi agent communication,” in *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2025.
 - [17] M. Yazgan, Q. Wu, I. Hamdard, S. Li, and J. M. Zoellner, “Slimcomm: Doppler-guided sparse queries for bandwidth-efficient cooperative 3-d perception,” in *ICCV Drive2X Workshop*, 2025.
 - [18] C. Li, R. Lin, A. Ji, and D. Xu, “Coanchor: Robust collaborative perception under spatio-temporal misalignment via object-level anchors,” *arXiv preprint arXiv:2608.21055*, 2026.